Jenkins Upgrade Guide (Ubuntu EC2 + APT Installation)

This document provides a complete, safe, and production-ready process to upgrade Jenkins from an older version (e.g., 2.504.1) to the latest LTS version on an Ubuntu EC2 instance.

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1. Overview

This guide covers:

- Pre-checks
- Full backup steps
- Adding Jenkins LTS repository
- Upgrading Jenkins safely
- Plugin updates
- Rollback procedure
- Post-upgrade validation

Environment:

- **Ubuntu EC2 instance**
- **Jenkins installed via APT**
- **Java 21 LTS installed**

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2. Pre-Upgrade Checks

Before upgrading, verify your Java version:

```
java -version
```

Ensure Java is **17 or above**. Java 21 LTS is recommended and fully supported.

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3. Stop Jenkins & Create Full Backup

Backup Jenkins home and configuration files.

3.1 Stop Jenkins

```
sudo systemctl stop jenkins
sudo systemctl status jenkins --no-pager
```

Expected: Jenkins should show `inactive (dead)`.

3.2 Backup Jenkins Home Directory

```
BACKUP_DIR="/var/backups/jenkins_backup_$(date +%F_%H-%M-%S)"
sudo mkdir -p "$BACKUP_DIR"
sudo cp -a /var/lib/jenkins "$BACKUP_DIR"/jenkins_home_copy
```

This backup includes:

- Jobs
- Plugins
- Credentials
- Build history
- Secrets
- Global config

3.3 Create a Compressed Backup (Recommended)

```
sudo tar -C /var/backups -czvf "${BACKUP_DIR}.tar.gz" "$(basename
"$BACKUP_DIR")"
```

3.4 Backup Jenkins Configuration Files

```
sudo cp -a /etc/default/jenkins "$BACKUP_DIR"/ || true
sudo cp -a /etc/init.d/jenkins "$BACKUP_DIR"/ || true
```

3.5 Restart Jenkins After Backup

```
sudo systemctl start jenkins
sudo systemctl status jenkins --no-pager
```

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4. Add the Jenkins LTS Repository

Add the official stable Jenkins repository.

4.1 Add Jenkins Key

```
wget -q -O - https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo
tee
  /usr/share/keyrings/jenkins-keyring.asc > /dev/null
```

4.2 Add Stable Repo

```
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]
  https://pkg.jenkins.io/debian-stable binary/ |
  sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

4.3 Update Package Index

```
sudo apt update
```

Ensure the Jenkins repo loads without errors.

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5. Upgrade Jenkins to Latest LTS Version

Run:

```
sudo apt install jenkins -y
```

This automatically:

- Upgrades Jenkins core
- Keeps jobs, plugins, data, configs intact
- Restarts Jenkins

You may see a kernel upgrade warning — this is unrelated to Jenkins.

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6. Verify Jenkins Status

After installation:

```
sudo systemctl status jenkins --no-pager
```

Expected:

```
Active: active (running)
Jenkins is fully up and running
```

If not running, check logs:

```
sudo journalctl -u jenkins -n 200 --no-pager
```

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7. Update Plugins (Required After Core Upgrade)

Inside Jenkins UI:

1. **Manage Jenkins**
2. **Plugins** → **Updates**
3. Click **Select All**
4. Click **Download now and install after restart**

This ensures full compatibility with the new Jenkins core.

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8. Rollback Procedure (Instant Restore)

If anything breaks, restore your backup:

8.1 Stop Jenkins

```
sudo systemctl stop jenkins
```

8.2 Restore Jenkins Home

```
sudo rm -rf /var/lib/jenkins
sudo cp -r /var/backups/jenkins_backup_YYYY-MM-DD_HH-MM-SS/jenkins_home_copy /
var/lib/jenkins
```

8.3 Start Jenkins

```
sudo systemctl start jenkins
```

Rollback completes in 2 minutes.

Document Complete

This guide provides a safe, repeatable, production-grade Jenkins upgrade procedure for Ubuntu EC2 instances.